



Manajemen Proyek Sistem Informasi

PROJECT HUMAN RESOURCE MANAGEMENT

S1 SISTEM INFORMASI

FAKULTAS REKAYASA INDUSTRI

TELKOM UNIVERSITY





Learning Objectives

- Explain the **importance of good human resource management on projects**, including the current state and future implications of the global IT workforce
- Define project human resource management and **understand its processes**
- Summarize key concepts for managing people by **understanding the theories of Abraham Maslow, Frederick Herzberg, David McClelland, and Douglas McGregor on motivation**, **H. J. Thamhain and D. L. Wilemon on influencing workers**, and **Stephen Covey on how people and teams** can become more effective
- Understand important issues involved in project staff acquisition and explain the concepts of resource assignments, resource loading, and resource leveling



Learning Objectives (Cont)

- Discuss human resource planning and be able to create a human resource plan, project organizational chart, responsibility assignment matrix, and resource histogram
- Understand important issues involved in project staff acquisition and explain the concepts of resource assignments, resource loading, and resource leveling
- Assist in team development with training, team-building activities, and reward systems
- Explain and apply several tools and techniques to help manage a project team and summarize general advice on managing teams
- Describe how project management software can assist in project human resource management



The Importance of Human Resource Management

- Many corporate executives have said, “People are our most important asset”
- People determine the success and failure of organizations and projects



Implications for the Future of IT Human Resource Management

- Proactive organizations are addressing workforce needs by:
 - Improving **benefits**
 - Redefining **work hours and incentives**
 - Finding **future workers**



What is Project Human Resource Management?

- Making the most effective use of the people involved with a project
- Processes include:
 1. **Planning human resource management:** identifying and documenting project roles, responsibilities, and reporting relationships
 2. **Acquiring the project team:** getting the needed personnel assigned to and working on the project
 3. **Developing the project team:** building individual and group skills to enhance project performance
 4. **Managing the project team:** tracking team member performance, motivating team members, providing timely feedback, resolving issues and conflicts, and coordinating changes to help enhance project performance



Project Human Resource Management Summary

Planning

Process: **Plan human resource management**

Output: Human resource plan

Executing

Process: **Acquire project team**

Outputs: Project staff assignments, resource calendars, project management plan updates

Process: **Develop project team**

Outputs: Team performance assessments, enterprise environmental factors updates

Monitoring and Controlling

Process: **Manage project team**

Outputs: Change requests, project management plan updates, project documents updates, enterprise environmental factors updates, and organizational process assets updates

Project Start

Project Finish



Keys to Managing People

- Psychologists and management theorists have devoted much **research and thought to the field of managing people** at work
- **Important areas** related to project management include:
 - Motivation theories
 - Influence and power
 - Effectiveness



Intrinsic and Extrinsic Motivation

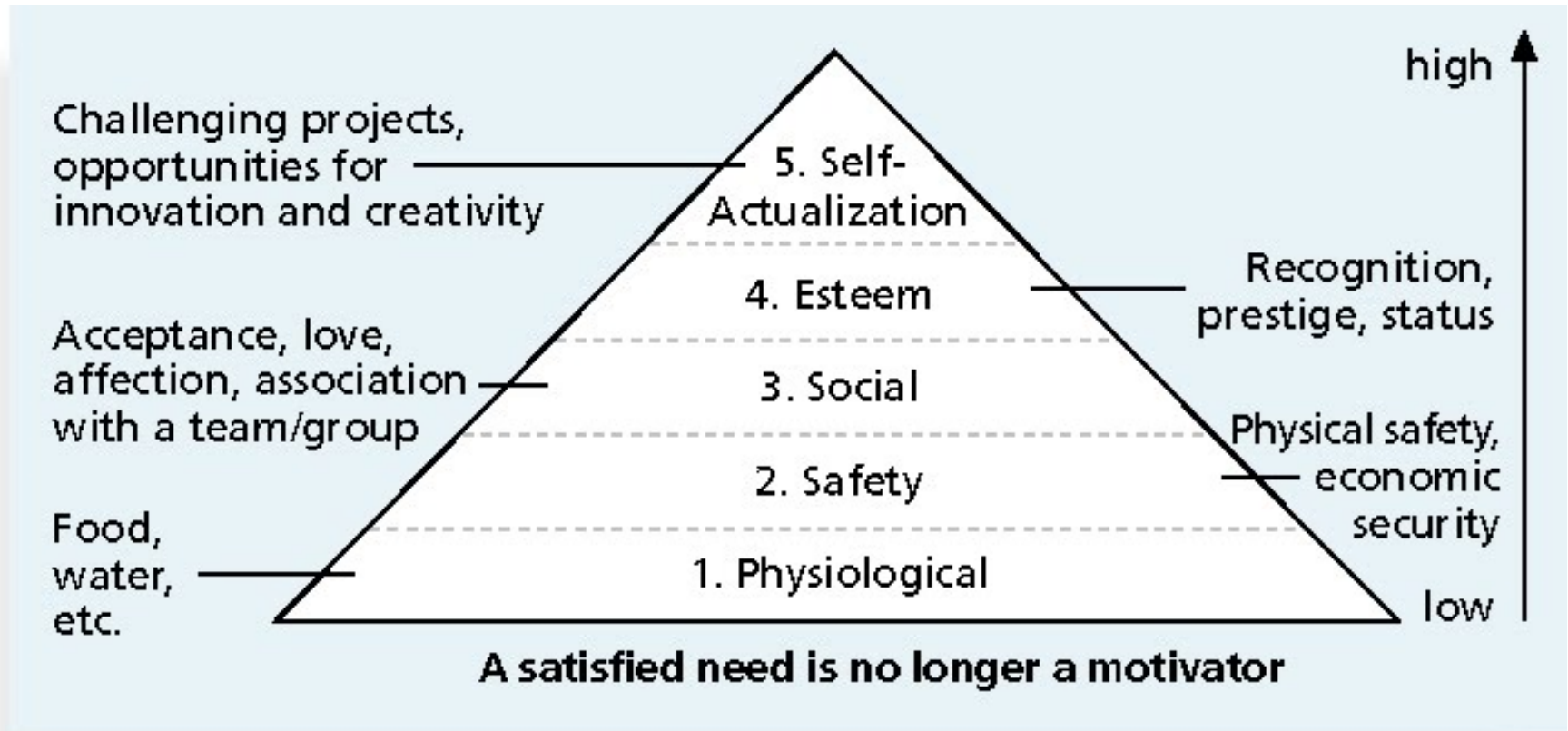
- **Intrinsic motivation** causes people to participate in an activity for their own enjoyment
- **Extrinsic motivation** causes people to do something for a reward or to avoid a penalty
- For example:
 - some children take piano lessons for **intrinsic motivation** (they enjoy it)
 - while others take them for **extrinsic motivation** (to get a reward or avoid punishment)



Maslow's Hierarchy of Needs

- Abraham Maslow argued that humans possess unique qualities that enable them to make independent choices, thus giving them control of their destiny
- Maslow developed a **hierarchy of needs** which states that **people's behaviors are guided or motivated by a sequence of needs**

Maslow's Hierarchy of Needs





Herzberg's Motivational and Hygiene Factors

- Frederick Herzberg wrote several famous books and articles about worker motivation; he distinguished between:
 1. **Motivational factors:** achievement, recognition, the work itself, responsibility, advancement, and growth, which produce job satisfaction
 2. **Hygiene factors:** cause dissatisfaction if not present, but do not motivate workers to do more; examples include larger salaries, more supervision, and a more attractive work environment



Examples of Herzberg's Hygiene Factors and Motivators

HYGIENE FACTORS	MOTIVATORS
Larger salaries	Achievement
More supervision	Recognition
More attractive work environment	Work itself
Computer or other required equipment	Responsibility
Health benefits	Advancement
Training	Growth





McClelland's **Acquired-Needs** Theory

- Specific needs are acquired or learned over time and shaped by life experiences, including:
 - **Achievement** (nAch): achievers like challenging projects with achievable goals and lots of feedback
 - **Affiliation** (nAff): people with high nAff desire **harmonious relationships** and **need to feel accepted by others**, so managers should try to create a cooperative work environment for them
 - **Power** (nPow): people with a need for power desire either personal power (not good) or institutional power (good for the organization); provide institutional power seekers with management opportunities



McGregor's Theory X and Y

- Douglas McGregor popularized the **human relations approach to management** in the 1960s
- **Theory X**: assumes **workers dislike and avoid work**, so managers must use coercion, threats, and various control schemes to get workers to meet objectives
- **Theory Y**: assumes **individuals consider work as natural as play or rest** and enjoy the satisfaction of esteem and self-actualization needs
- **Theory Z**: introduced in 1981 by William Ouchi and is **based on the Japanese approach to motivating workers**, emphasizing trust, quality, collective decision making, and cultural values



Thamhain and Wilemon's **Ways to Have Influence on Projects**

1. **Authority**: the legitimate **hierarchical right to issue orders**
2. **Assignment**: the project manager's perceived **ability to influence a worker's later work assignments**
3. **Budget**: the project manager's perceived **ability to authorize others' use of discretionary funds**
4. **Promotion**: the **ability to improve a worker's position**
5. **Money**: the **ability to increase a worker's pay and benefits**





Thamhain and Wilemon's **Ways to Have Influence on Projects** (continued)

6. **Penalty**: the project manager's **ability to cause punishment**
7. **Work challenge**: the **ability to assign work that capitalizes on a worker's enjoyment** of doing a particular task
8. **Expertise**: the project manager's perceived **special knowledge that others deem important**
9. **Friendship**: the **ability to establish friendly personal relationships** between the project manager and others



Ways to Influence that Help and Hurt Projects

- Projects are **more likely to succeed** when project managers influence with:
 - Expertise
 - Work challenge
- Projects are **more likely to fail** when project managers rely too heavily on:
 - Authority
 - Money
 - Penalty



Power

- **Power** is the potential ability to influence behavior to get people to do things they would not otherwise do
- Types of power include:
 1. Coercive
 2. Legitimate
 3. Expert
 4. Reward
 5. Referent



Covey and Improving Effectiveness

- Project managers can apply **Covey's 7 habits to improve effectiveness** on projects
 1. Be proactive
 2. Begin with the end in mind
 3. Put first things first
 4. Think win/win
 5. Seek first to understand, then to be understood
 6. Synergize
 7. Sharpen the saw



Empathic Listening and Rapport

- Rapport is a **relation of harmony, conformity, accord, or affinity**
- **Good project managers** are **empathic listeners**; they listen with the intent to understand
- Before you can communicate with others, **you have to have rapport**, a **relation of harmony, conformity, accord, or affinity**
- **Mirroring** is the **matching of certain behaviors of the other person**, a technique to help establish rapport
- IT professionals need to **develop empathic listening** and other people **skills to improve relationships** with users and other stakeholders

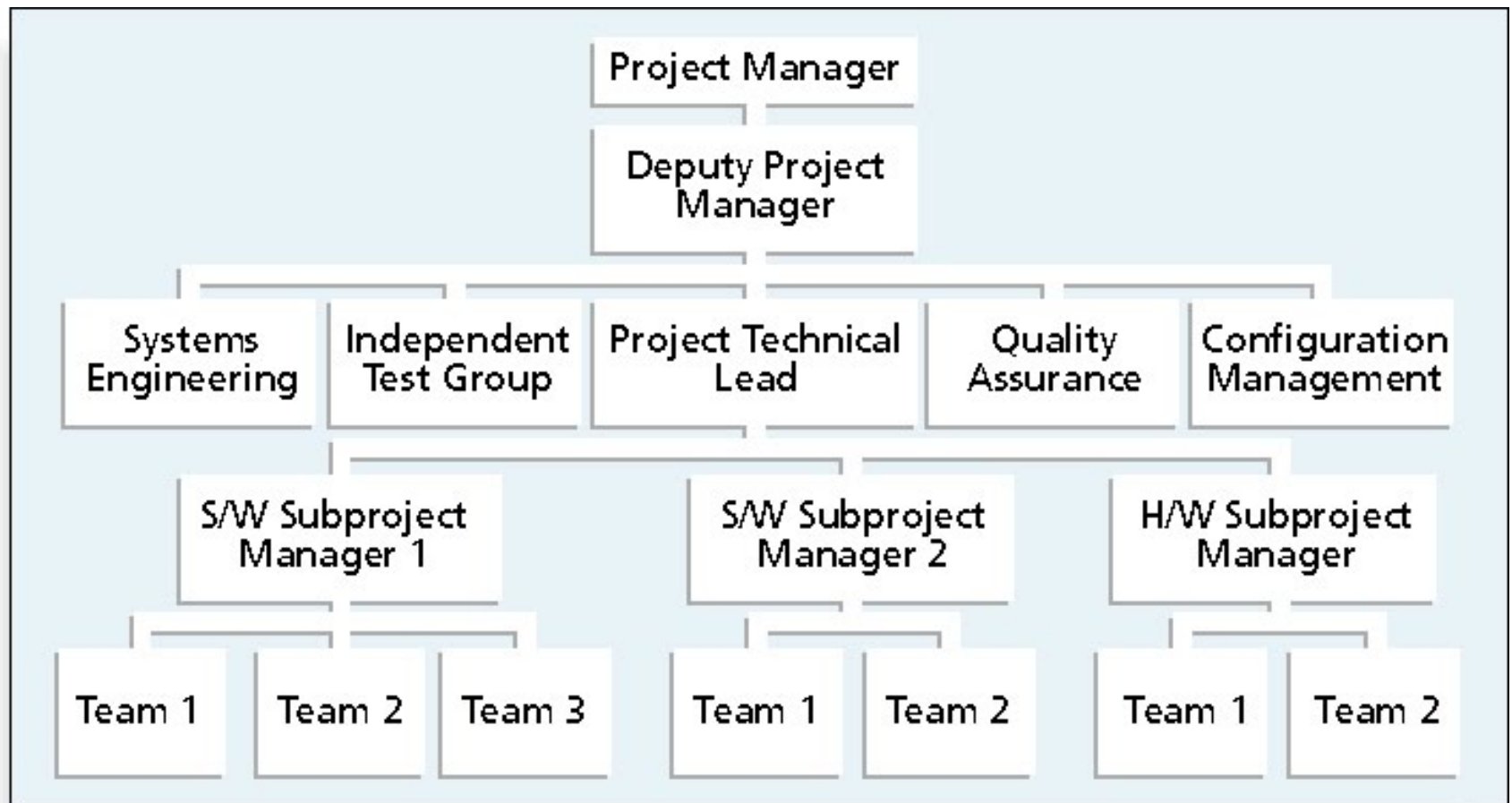


1. Developing the Human Resource Plan

- Involves identifying and documenting project roles, responsibilities, and reporting relationships
- Contents include:
 - Project organizational charts
 - Staffing management plan
 - Responsibility assignment matrixes
 - Resource histograms

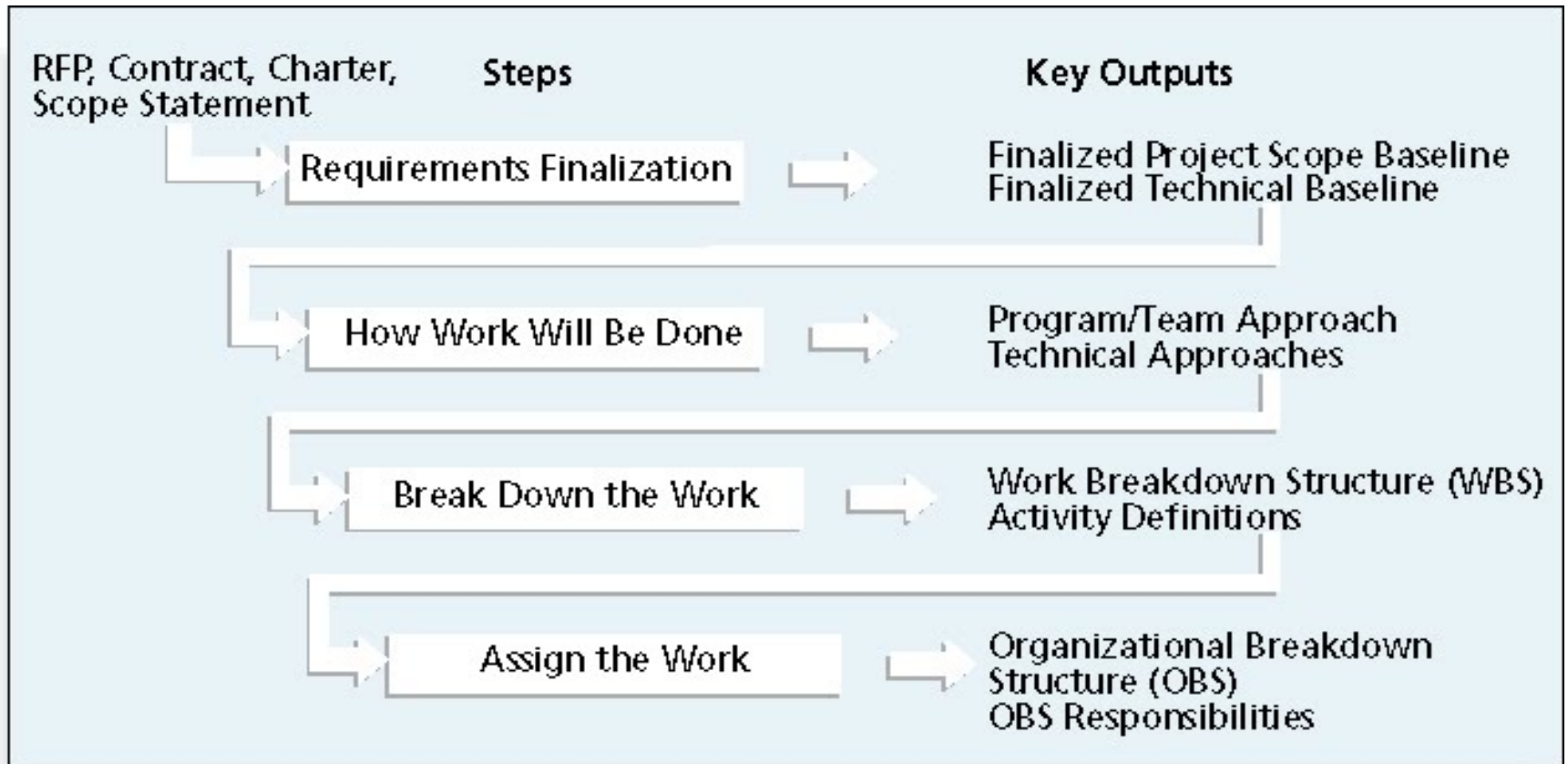


Sample Organizational Chart for a Large IT Project





Work Definition and Assignment Process





Responsibility Assignment Matrices

- A **responsibility assignment matrix (RAM)** is a **matrix that maps the work of the project as described in the WBS** to the people responsible for performing the work as described in the OBS
- Can be created in different ways to meet unique project needs



Sample Responsibility Assignment Matrix (RAM)

WBS activities →

OBS units ↓

	1.1.1	1.1.2	1.1.3	1.1.4	1.1.5	1.1.6	1.1.7	1.1.8
Systems Engineering	R	R P					R	
Software Development			R P					
Hardware Development				R P				
Test Engineering	P							
Quality Assurance					R P			
Configuration Management						R P		
Integrated Logistics Support							P	
Training								R P

R = Responsible organizational unit
P = Performing organizational unit





RAM Showing Stakeholder Roles

Items	Stakeholders				
	A	B	C	D	E
Unit Test	S	A	I	I	R
Integration Test	S	P	A	I	R
System Test	S	P	A	I	R
User Acceptance Test	S	P	I	A	R

A = Accountable

P = Participant

R = Review Required

I = Input Required

S = Sign-off Required



Sample RACI Chart

	Group A	Group B	Group C	Group D	Group E
Test Plans	R	A	C	C	I
Unit Test	C	I	R	A	I
Integration Test	A	R	I	C	C
System Test	I	C	A	I	R
User Acceptance Test	R	I	C	R	A

R = responsibility

A = accountability, only one A per task

C = consultation

I = informed

Note that some people reverse the definitions of responsible and accountable

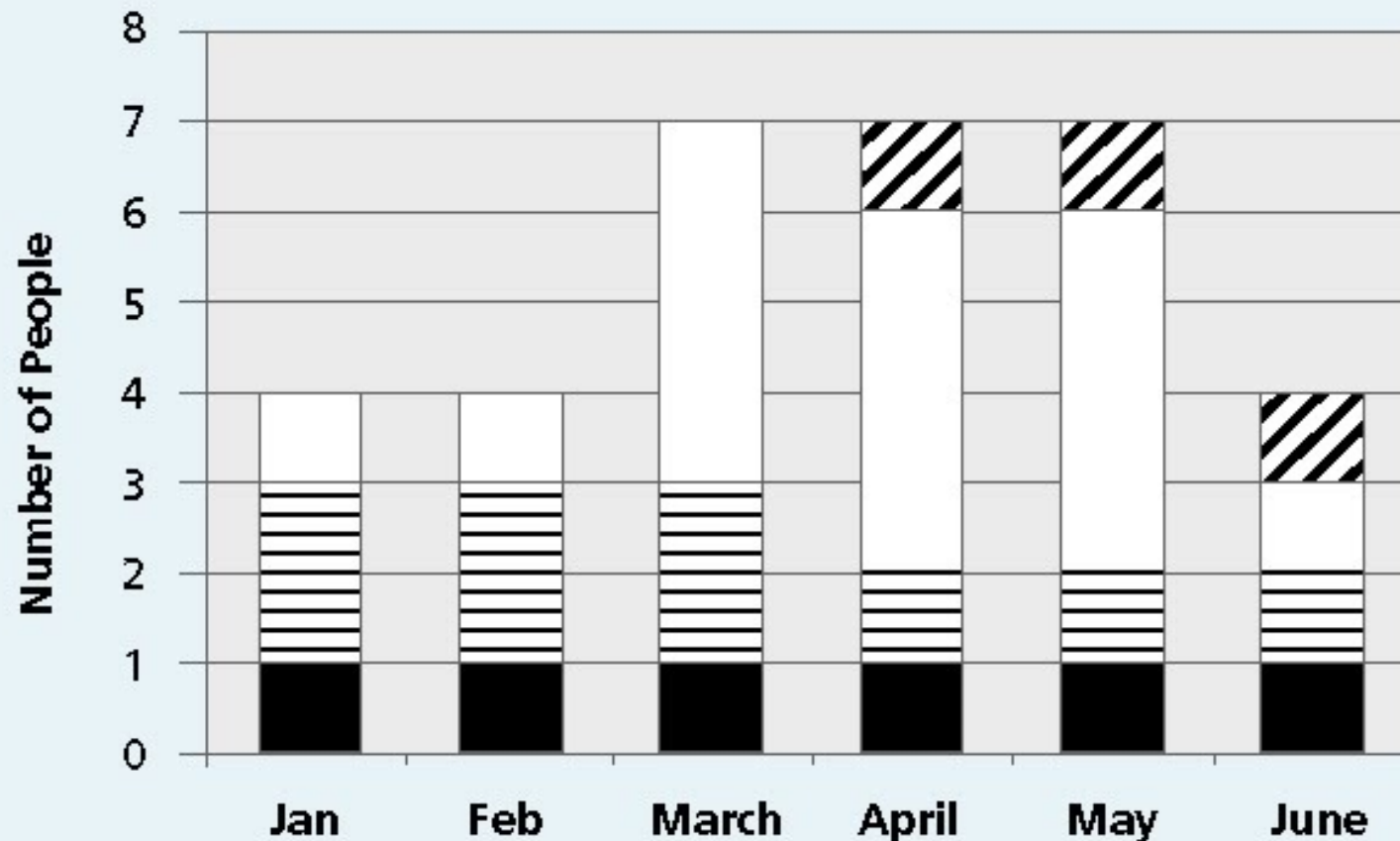


Staffing Management Plans and Resource Histograms

- A **staffing management plan** describes **when and how people will be added to and taken off the project team**
- A **resource histogram** is a **column chart that shows the number of resources assigned to a project over time**



Sample Resource Histogram



■ Managers ▨ Business analysts □ Programmers ▩ Technical writers



Plan Human Resource Management ITTO

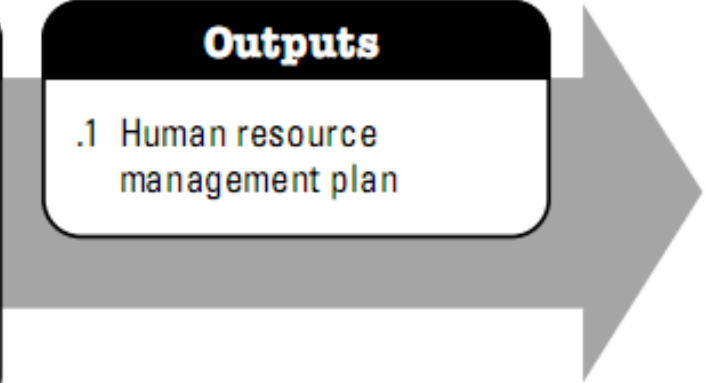
Inputs

- .1 Project management plan
- .2 Activity resource requirements
- .3 Enterprise environmental factors
- .4 Organizational process assets

Tools & Techniques

- .1 Organization charts and position descriptions
- .2 Networking
- .3 Organizational theory
- .4 Expert judgment
- .5 Meetings

Outputs

- .1 Human resource management plan
- 



2. Acquiring the Project Team

- Acquiring qualified people for teams is crucial
- The project manager who is the smartest person on the team has done a poor job of recruiting!
- It's important to assign the appropriate type and number of people to work on projects at the appropriate times



Resource Assignment

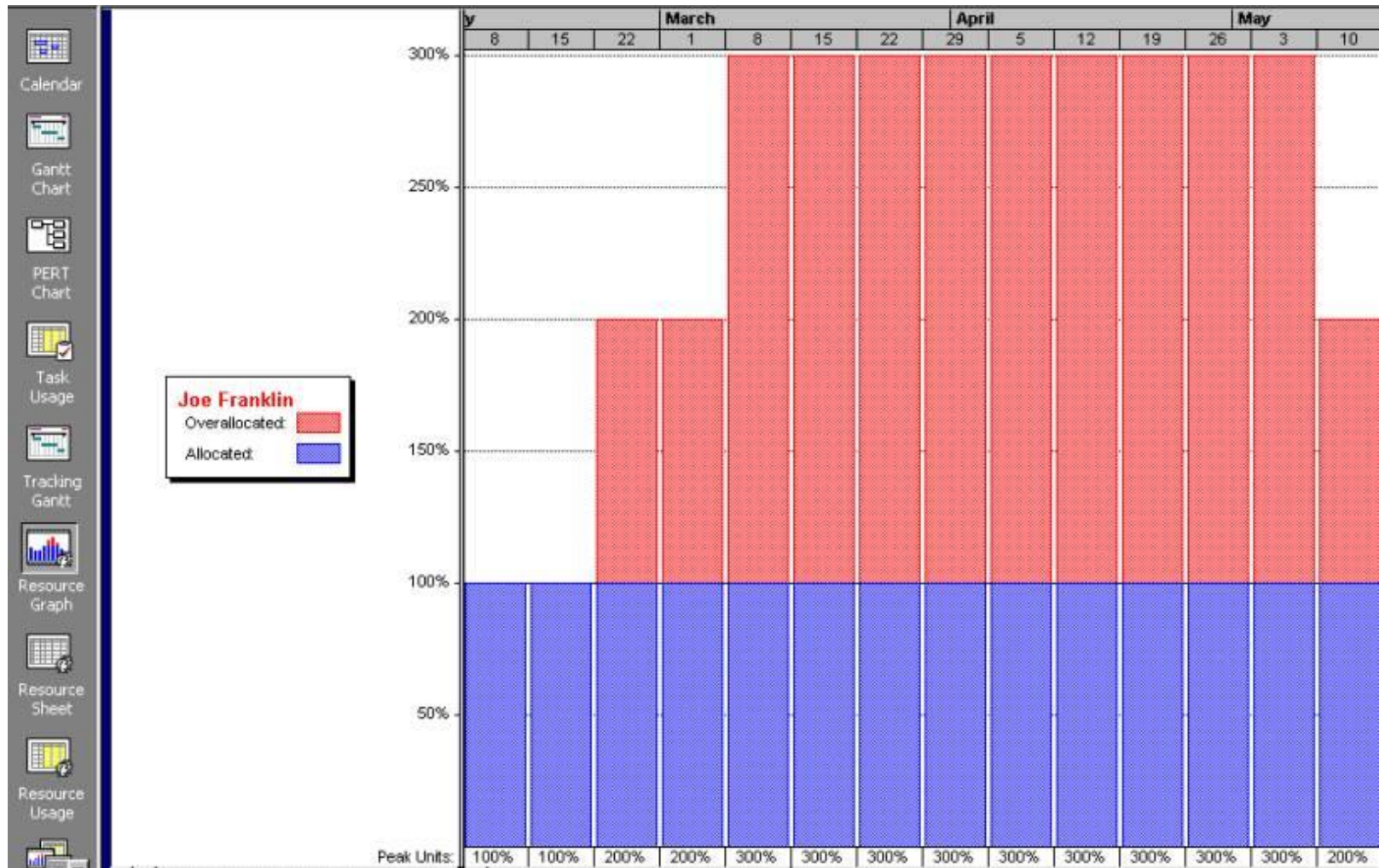
- Staffing plans and good hiring procedures are important, as are **incentives for recruiting** and **retention**
 - Some companies give their employees one dollar for every hour a new person they helped hire works
 - Some organizations **allow people to work from home as an incentive**
- Enrollment in U.S. computer science and engineering programs has dropped almost in half since 2000, and **one-third of U.S. workers will be over the age of 50 by 2010**
- CIO's researchers suggest that organizations rethink hiring practices and incentives to hire and retain IT talent



Resource Loading

- **Resource loading** refers to the amount of individual resources an existing schedule requires during specific time periods
- Helps project managers develop a general understanding of the demands a project will make on the organization's resources and individual people's schedules
- **Overallocation** means that more resources than are available are assigned to perform work at a given time

Sample Histogram Showing an Overallocated Individual

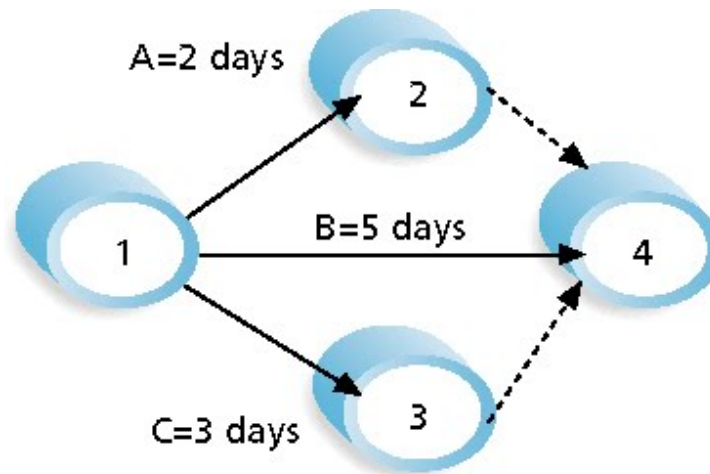




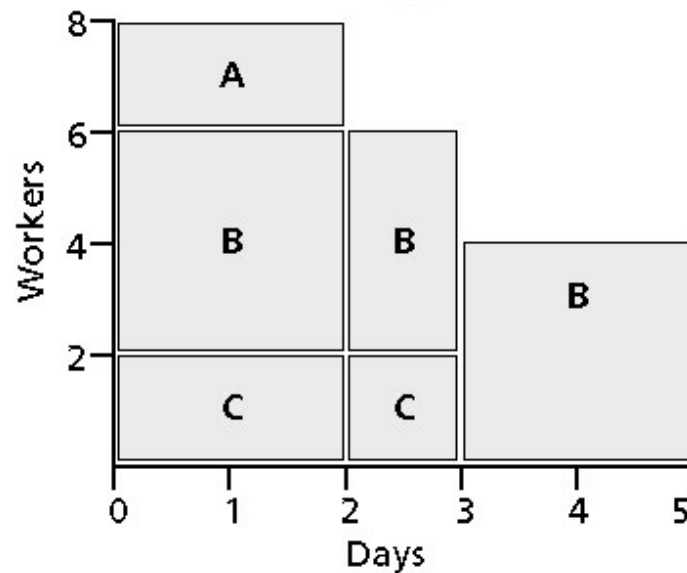
Resource Leveling

- **Resource leveling** is a **technique for resolving resource conflicts** by delaying tasks
- The **main purpose of resource leveling** is to **create a smoother distribution of resource usage** and **reduce overallocation**

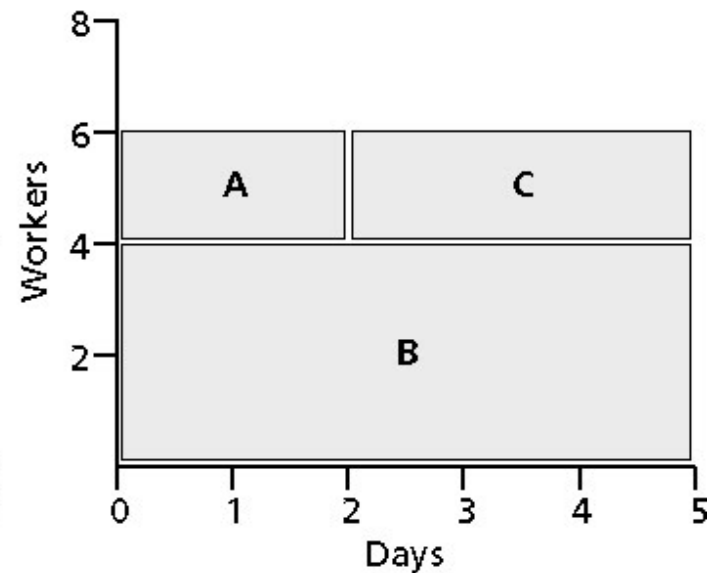
Resource Leveling Example



Project network with Activities A, B, and C and durations as shown. Activity A has 3 days of slack, and Activity C has 2 days of slack. Assume Activity A has 2 workers, B has 4 workers, and C has 2 workers.



Resource usage if all activities start on day one



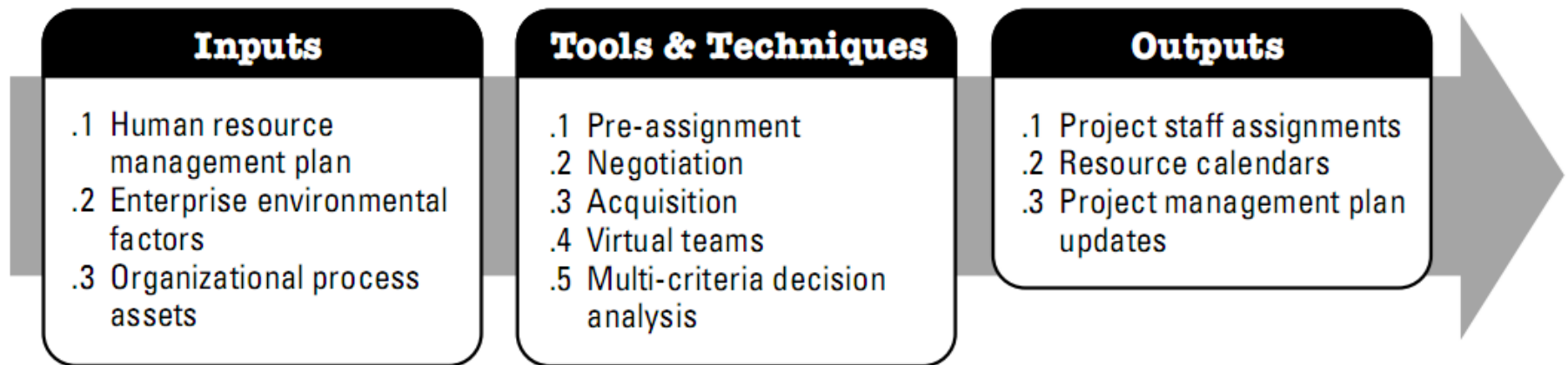
Resource usage if Activity C is delayed 2 days, its total slack



Benefits of Resource Leveling

- When resources are used on a more constant basis, they require less management
- It may enable project managers to use a just-in-time inventory type of policy for using subcontractors or other expensive resources
- It results in fewer problems for project personnel and accounting department
- It often improves morale

Acquire Project Team ITTO





3. Developing the Project Team

- The main **goal of team development** is **to help people work together more effectively** to improve project performance
- It takes teamwork to successfully complete most projects



Tuckman Model of Team Development

- Forming
- Storming
- Norming
- Performing
- Adjourning



Training

- **Training** can help people understand themselves, each other, and how to work better in teams
- **Team building** activities include:
 - Physical challenges
 - Psychological preference indicator tools





Meyers-Briggs Type Indicator (MBTI)

- **MBTI** is a popular **tool for determining personality preferences** and **helping teammates understand each other**
- **Four dimensions** include:
 1. Extrovert/Introvert (E/I)
 2. Sensation/Intuition (S/N)
 3. Thinking/Feeling (T/F)
 4. Judgment/Perception (J/P)
- NTs or **rational**s are **attracted to technology fields**
- IT people vary most from the general population in not being extroverted or sensing





Social Styles Profile

- People are perceived as **behaving primarily in one of four zones**, based on their **assertiveness and responsiveness**:
 1. Drivers
 2. Expressives
 3. Analyticals
 4. Amiables
- **People on opposite corners** (drivers and amiables, analyticals and expressives) **may have difficulties getting along**



DISC Profiles

- Also uses a **four-dimensional model of normal behavior**
 1. Dominance
 2. Influence
 3. Steadiness
 4. Compliance
- **People in opposite quadrants** can **have problems understanding each other**





The DISC Profile

It **Compliance (Blue)**

Data driven, risk averse,
concerned, works well alone,
prefers processes and procedures,
not very communicative or social

I **Dominance (Red)**

Direct, decisive, assertive,
outcome oriented, competitive,
self assured, takes control, has to
win

You **Steadiness (Green)**

Calm, sincere, sympathetic,
cooperative, cautious, conflict
averse, good listener, wants to
maintain stability

We **Influence (Yellow)**

Persuasive, optimistic,
outgoing, verbal, enthusiastic,
strives to win others over,
leadership through acclimation

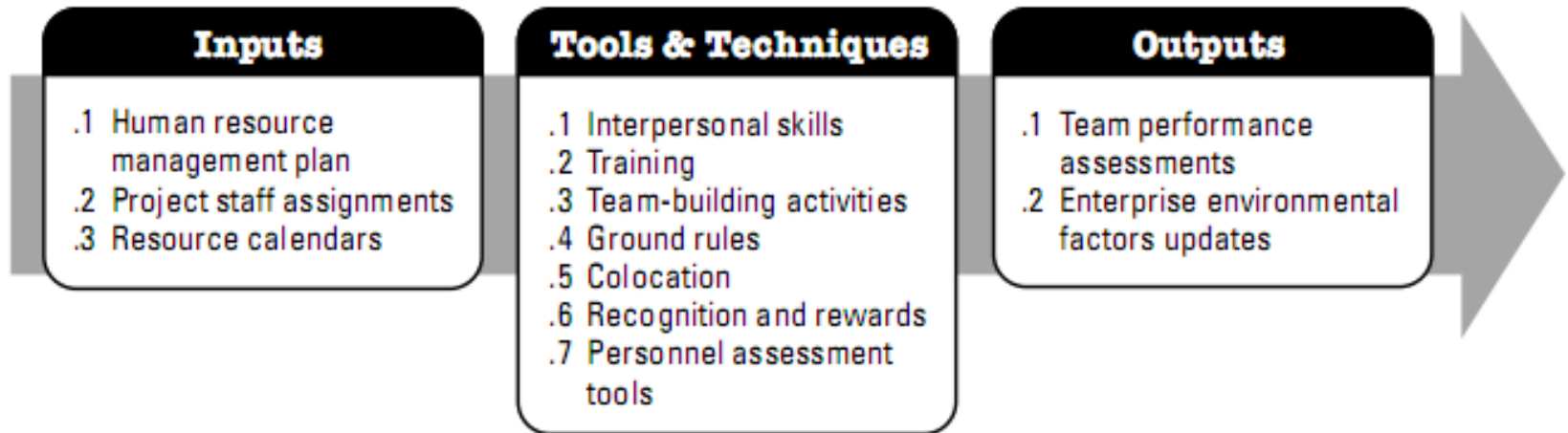




Reward and Recognition Systems

- Team-based reward and recognition systems can promote teamwork
- Focus on rewarding teams for achieving specific goals
- Allow time for team members to mentor and help each other to meet project goals and develop human resources

Develop Project Team ITTO

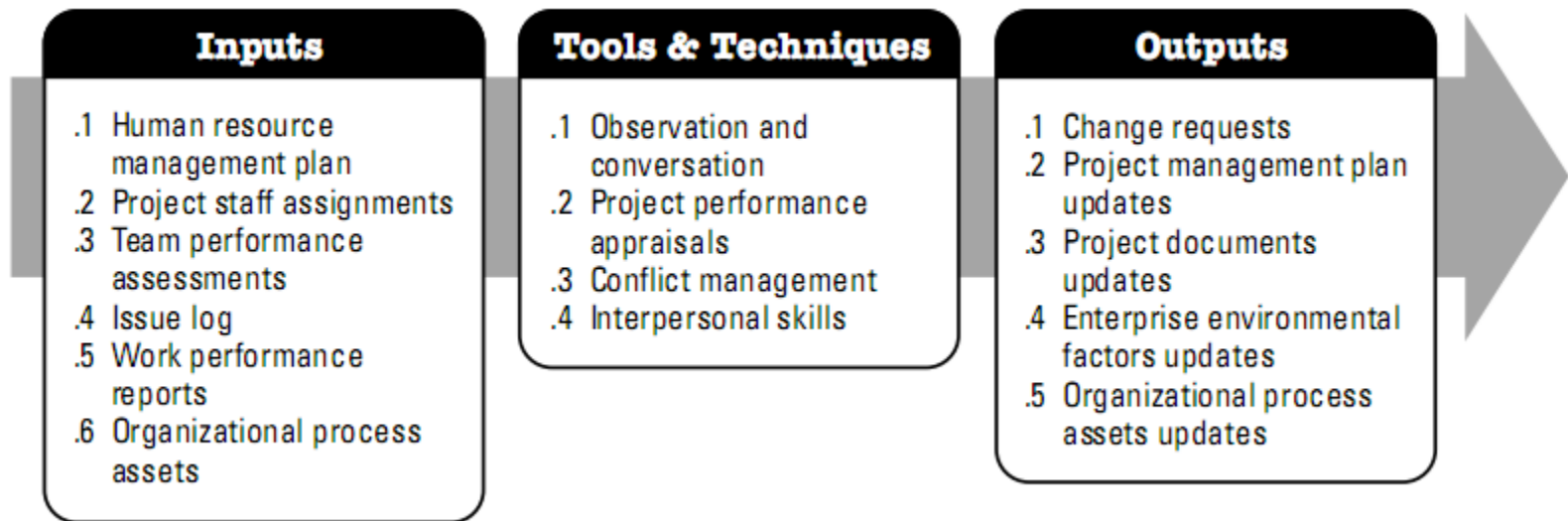




4. Managing the Project Team

- Project managers must **lead their teams** in **performing various project activities**
- After assessing team performance and related information, the project manager must decide:
 - If changes should be requested to the project
 - If corrective or preventive actions should be recommended
 - If updates are needed to the project management plan or organizational process assets

Manage Project Teams ITTO





Tools and Techniques for Managing Project Teams

- Observation and conversation
- Project performance appraisals
- Conflict management
- Issue logs
- Interpersonal skills





General Advice on Teams

- Be patient and kind with your team
- Fix the problem instead of blaming people
- Establish regular, effective meetings
- Allow time for teams to go through the basic team-building stages
- Limit the size of work teams to three to seven members



General Advice on Teams (continued)

- Plan some social activities to help project team members and other stakeholders get to know each other better
- Stress team identity
- Nurture team members and encourage them to help each other
- Take additional actions to work with virtual team members



Five Dysfunctions of a Team

- Patrick Lencioni, author of several books on teams, says that “Teamwork remains the one sustainable competitive advantage that has been largely untapped”*
 - The five dysfunctions of teams are:
 - Absence of trust
 - Fear of conflict
 - Lack of commitment
 - Avoidance of accountability
 - Inattention to results

*Lencioni, Patrick, “Overcoming the Five Dysfunctions of a Team,” Jossey-Bass: San Francisco, CA (2005), p. 3.



Using Software to Assist in Human Resource Management

- Software can help in producing RAM and resource histograms
- Project management software includes several features related to human resource management such as:
 - Assigning resources
 - Identifying potential resource shortages or underutilization
 - Leveling resources





Benefit Using Software to Assist in Human Resource Management

- Keep track of resources through stored information and reports on resource assignments.
- Identify potential resource shortages that could force a project to miss scheduled deadlines and possibly extend the duration of a project.
- Identify underutilized resources and reassign them, which may enable you to shorten a project's schedule and reduce costs.
- Use automated leveling to make level resources easier to manage.



Project Resource Management Involves Much More Than Using Software

- Project managers must:
 - Treat people with consideration and respect
 - Understand what motivates them
 - Communicate carefully with them
- Focus on your goal of enabling project team members to deliver their best work





Summary

- Project human resource management includes the **processes required to make the most effective use of the people involved** with a project
- Main processes include:
 1. Plan human resource management
 2. Acquire project team
 3. Develop project team
 4. Manage project team





References

1. Kathy Schwalbe, **Managing Information Technology Projects 7th Edition**, *Course Technology, Cengage Learning*, 2014
2. A Guide to the Project Management Body of Knowledge: **PMBOK Guide 6th Edition**, *Project Management Institute*, 2017